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PROJECT TITLE: AI-POWERED INTELLIGENT AUTOMATED TOLL COLLECTION SYSTEM WITH LICENSE PLATE RECOGNITION (LPR) AND IOT INTEGRATION FOR PLUS HIGHWAYS

ABSTRACT

Managing toll collection efficiently to ensure smooth traffic flow has long been a critical challenge, especially during peak hours. The current legacy system suffers from inefficiencies, such as traffic congestion caused by slow payment processes via cash, Touch 'n Go (TNG) cards, and RFID tags. The proposed system leverages AI technologies like image processing, computer vision, and cloud computing for data management. By consolidating these technologies, toll operational efficiency will be enhanced, leading to reduced congestion and improved user experience. This project stands out by integrating AI-powered vehicle number plate recognition, automated toll payments, and real-time traffic monitoring and management, creating a smarter, more scalable solution for highway toll management.

PROBLEM

The existing toll collection systems on the PLUS Highways, which form a critical backbone of Malaysia's transportation network, face significant challenges in delivering an efficient and seamless experience for both commuters and operators. These inefficiencies manifest in outdated payment processes, technical failures, and inadequate traffic management capabilities, resulting in widespread commuter dissatisfaction, operational inefficiencies, and environmental impacts.

This problem is worth solving because it directly impacts 1.6 million of daily commuters (Alias, 2023). Existing inefficiency exacerbate traffic congestion in high-traffic urban areas such as Penang (MalayMail, 2023), Kuala Lumpur, Selangor, Johor, and Malacca (MalaysiaKini, 2024). This wastes valuable time and fuel while contributing to economic and environmental costs. As the saying goes, "Time is gold". For instance, traffic delays lead to wasted time (Akhtar, 2024) and fuel (Jereb, 2018), increasing transportation costs for individuals and businesses. Prolonged congestion worsens air pollution, contradicting environmental sustainability goals (Shuhua Xy && Chuanwang Sun && Nian Liu, 2024).

The reliance on manual cash payments, RFID tags, and Touch 'n Go (TNG) cards introduces several technical vulnerabilities. Issues include RFID reader incompatibilities, TNG card malfunctions, and the wear-and-tear of mechanical systems in cash-handling facilities such as cash registers and coin dispensers. These problems not only disrupt operations but also escalate maintenance costs, lowering the Return on

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Investment (ROI) for toll operators. PLUS Malaysia, for instance, faces financial challenges, including an outstanding debt of RM 30.2 billion (Alifah Zainudin && Dashveenjit Kaur, 2019).

Last but not least, the absence of real-time traffic monitoring, management and advanced tolling technologies limits the ability of operators to proactively address congestion and optimise road conditions. This lack of adaptability hinders their capacity to analyse operational data such as vehicle count effectively, further reducing system reliability and revenue potential.

In conclusion, the over-reliance on outdated technologies like RFID and TNG cards, combined with the inefficiencies of manual cash payment systems, leads to significant delays, increased operational costs, and environmental harm. These challenges highlight the urgent need for a modernized toll collection system leveraging AI-driven predictive analytics and real-time monitoring to enhance efficiency, reduce costs, and align with sustainability goals.

SOLUTION

The AI-powered Intelligent Automated Toll Collection System is designed to address the inefficiencies of current toll systems, aiming to reduce congestion, improve toll efficiency, and enhance the commuter experience. The system leverages Firebase for centralized payment processing and real-time data management, ensuring secure user sign-ins through Firebase Authentication and utilizing Firebase Cloud Functions to handle toll payments and charge processing. Integration with secure payment gateways, such as Stripe, PayPal, or Razorpay, facilitates seamless transactions using various digital payment methods, including banking cards.

The core functionality of the system involves linking user accounts to vehicles by associating each user with their vehicle's license plate. This enables automated toll deductions through contactless payments. The system uses OpenCV for vehicle license plate detection and Convolutional Neural Networks (CNN) for Optical Character Recognition (OCR) to match detected license plates with user accounts stored in Firebase. Upon successful recognition, toll payments are automatically triggered. All user and vehicle data are securely stored in Firebase, allowing for real-time syncing and processing.

For real-time traffic management, the system integrates advanced technologies like YoloV8, a deep learning model for vehicle counting and traffic flow calculation. Additionally, the system sends congestion alerts to commuters when necessary, optimizing the toll experience. Advanced mathematical formulas

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are also used to calculate traffic flow, road capacity, and capacity utilization, providing insights into traffic efficiency and identifying potential congestion areas.

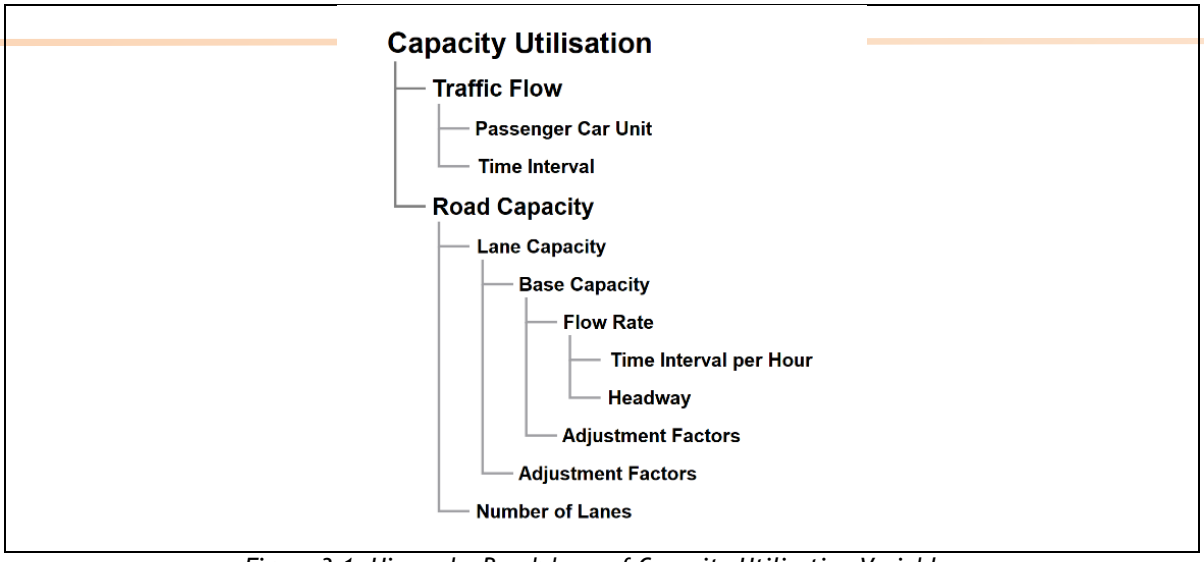

Figure 3.1: Hierarchy Breakdown of Capacity Utilisation Variables

To elaborate more on Figure 3.1, Flow Rate refers to the number of vehicles passing a specific point on the road per hour under ideal free-flowing traffic conditions, calculated as Flow Rate (PCU/hr) = (Time Interval per Hour (s)) / (Headway (s/PCU)) × 60, where Time Interval per Hour is typically 3600 seconds (1 hour), and Headway is the time gap between two consecutive vehicles. Base Capacity represents the theoretical maximum number of vehicles a lane can accommodate under ideal conditions, calculated as Base Capacity (PCU/hr) = Flow Rate × Adjustment Factors, with Adjustment Factors accounting for real-world conditions such as traffic volume, vehicle type, road design, traffic speed variations, and weather. Lane Capacity is the maximum number of vehicles that can pass through a specific section of a single lane in a given time period, calculated as Lane Capacity (PCU/hr/lane) = Base Capacity × Adjustment Factors. Road Capacity refers to the total capacity of all available lanes combined, calculated as Road Capacity (PCU/hr) = Lane Capacity × Number of Lanes. Traffic Flow is the actual number of vehicles passing a specific point on the road within a given time period, calculated as Traffic Flow (PCU/hr) = (Passenger Car Unit) / (Time Interval (min)) × 60. Capacity Utilization represents the ratio of actual traffic flow to maximum road capacity, expressed as a percentage, calculated as Capacity Utilization (%) = (Traffic Flow) / (Road Capacity) × 100, indicating how efficiently the road is being utilized.

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In addition, the system incorporates advanced weather-responsive features by utilizing data from integrated sensors, such as rain sensors and humidity sensors, to dynamically adjust speed limits and provide real-time alerts to users regarding adverse weather conditions like rain or fog. These enhancements not only improve safety but also optimize traffic flow during challenging weather scenarios.

Overall, the system is designed with distinct interfaces tailored to its targeted primary users. User-friendly mobile app is for commuters' personal use while comprehensive web-based dashboard is for administrators' professional use. The mobile app, optimized for ease of use on smartphones, is ideal for commuters to manage their accounts. On the other hand, the web-based dashboard is better suited for administrators who can leverage the convenience of computers to efficiently monitor traffic and manage toll operations.

For the development of the system, Flutter is going to be used for the mobile app frontend, which is written in Dart. Flutter allows the creation of a single codebase for Android, offering native performance and seamless integration with Firebase. It supports Firebase plugins for essential features like authentication, Firestore/Realtime Database, and Cloud Functions, making it an efficient choice for cross-platform development.

For the web dashboard, HTML, CSS, and JavaScript would be used for the frontend. As a popular JavaScript library for building interactive UIs, React.js offers excellent support for Firebase integration. For the backend, Firebase Cloud Functions is an ideal choice, written in JavaScript. It enables the implementation of server-side logic without the need to manage a separate backend server and integrates directly with Firebase services like Firestore, Realtime Database, and Authentication, simplifying development and ensuring scalability.

The user-side mobile app offers commuters a range of features to enhance convenience and functionality. Users can manage their accounts, link multiple vehicles, check real-time balances, and receive notifications for low balances or upcoming toll payments. The app provides detailed toll payment records, including date, time, location, and amount, with a downloadable transaction history for tax reporting. Additional features could include account management, enabling users to update personal details, view payment history, and track toll usage seamlessly. A toll usage history and reports section would give users insights into their spending patterns, useful for personal budgeting or reporting purposes. Notifications and alerts could keep users informed about system maintenance, traffic updates, weather updates, and low balances through push notifications for kind reminders. The app could also support prepaid balance recharges through digital payment option for automatic toll deductions. To enhance user engagement, rewards that are offering discounts will be introduced. Support and feedback features would ensure users can easily report issues or provide suggestions for improvement. An SOS (Save our souls) emergency

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assistance feature provides quick access to help during breakdowns or accidents. Lastly, vehicle management would allow users to link and manage multiple vehicles within a single account, simplifying toll tracking and history retrieval. These modules together create a comprehensive and user-friendly experience, addressing the diverse needs of commuters.

The admin interface should feature real-time traffic monitoring, enabling administrators to track toll booth activity, flow rates, and congestion levels to optimize toll collection. Advanced reporting tools can generate detailed insights into toll revenue to support informed decision-making and forecasting. User management capabilities would allow admins to handle account-related tasks, such as blocking suspicious activities, resolving disputes, and making account adjustments. Customisable toll rate functionality should enable dynamic adjustments based on the decisions of PLUS management teams. This is because there may some toll fare exemption subsidies by the government for toll fare for Malaysian welfare purposes sometimes (TheSun, 2024). An analytics dashboard with data visualizations like charts or graphs would make it easier for administrators to analyse trends, including traffic flow. Additionally, the web-based interface should generate detailed revenue reports, facilitate toll rate adjustments, and log recognised license plates with associated payment details, timestamps, and vehicle information, streamlining operational efficiency and system oversight.

The entire system is built to be scalable, secure, and efficient, leveraging modern cloud infrastructure via Firebase and AI-driven technologies for license plate recognition and traffic monitoring. This ensures a streamlined, contactless toll experience that reduces operational costs and enhances traffic flow management for the future.

TARGET MARKET

The proposed system is designed to serve two primary target market segments: commuters and toll management officers.

PLUS Highways handle approximately 1.6 million vehicle daily, positioning commuters as the largest user base (Alias, 2023). Targeted user base is substantial, with a focus on urban commuters, frequent travelers, and commercial transport operators.

Firstly, urban commuters are individuals who travel daily between cities for work, education, or other purposes. Urbanisation trends in Malaysia particularly in states such as Penang (MalayMail, 2023), Kuala Lumpur, Selangor, Johor, and Malacca (MalaysiaKini, 2024), will have a large pool of daily highway users. High majority of them will benefit from reduced travel times and streamlined toll collection.

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Moreover, frequent travelers are highway users taking intercity trips for personal or professional purposes. Undeniably, they prioritise convenience and efficient toll payment systems, particularly during holiday seasons when intercity travel increases.

Commercial transport operators, including logistics companies and freight movers, that rely on the highways for freight movement. Malaysia's logistics and freight transport market, estimated to grow at a compound annual growth rate (CAGR) of 8.72% by 2027 (MordorIntelligence, 2024). Examples of potential logistics companies are DHL Group, FedEx, DB Shenker, J&T Express, and MMC Corporation Berhad (MordorIntelligence, 2024). This indicates a significant opportunity for targeted services. Features like automated payment and congestion detection will help this segment minimise delivery delays, improve productivity, maximise cost efficiency, and enhance their customer satisfactions.

Toll management officers are a crucial audience for the proposed AI-powered toll system. This group includes plaza operators, who manage daily toll booth operations, and centralized monitoring teams, who oversee traffic and operational systems across the entire PLUS highway network, which encompasses 94 toll plazas (PLUS, 2024). PLUS Malaysia currently employs approximately 3,000 personnel, many of whom are involved in toll management and operations (PitchBook, 2024).

For plaza operators, the system's automated payment and real-time congestion detection reduce manual workload, making toll collection more efficient. This allows them to redirect their efforts to higher-priority tasks, such as customer assistance and incident management. Centralized monitoring teams, on the other hand, benefit from advanced analytics and traffic monitoring tools, enabling quicker decision-making, better resource allocation, and improved response times to traffic issues.

COMPETITION/CONTRIBUTION

The primary competitor for my proposed AI-powered Intelligent Automated Toll Collection System is the Dart Charge toll system in the United Kingdom. Automatic Number Plate Recognition (ANPR) is utilised by Dart Charge payment system in order to process toll payments at the Dartford Crossing, where vehicle pass through toll points without stopping (RapidVM, 2024). Their system employs camera to capture vehicle license plates. Also, their tolls are automatically deducted from pre-registered accounts. Similar to the proposed system, Dart Charge has successfully eliminated the traditional toll booths, reduced congestion and improving traffic flow (Gov.UK, 2024).

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However, Dart Charge has certain limitations that could be addressed by the proposed system. For example, it primarily focuses on passive toll collection and does not integrate real-time weather and traffic data. While Dart Charge does use variable message signs (VMS) or so-called “LED traffic signboards” to display speed limits and traffic updates, these are mainly adjusted manually, rather than dynamically responding to weather. On top of that, the lack of congestion level detection in Dart Charge means that their commuters may not be able to plan their journeys for heavy traffic congestion purposes (Gov.UK, 2024).

The proposed AI-powered toll system is a transformative solution that not only improves operational efficiency but also addresses significant community problems. By leveraging advanced technologies like real-time data processing, computer vision, and OCR-based license plate recognition, the system is designed to be faster, cheaper, and better than traditional toll collection methods. It processes toll payments seamlessly, detects congestion levels in real time, and enhances road safety, providing a meaningful societal impact by improving traffic flow and reducing environmental emissions.

Compared to competitors, the proposed solution stands out for its ability to automate toll operations, cutting down operational costs and minimising reliance on toll booth operators. This leads to substantial cost savings for the PLUS company while offering commuters a smoother, more reliable travel experience. The system's scalability and integration with existing infrastructure ensure its long-term value.

In solving societal problems, the system enhances convenience for commuters, reduces vehicle idling and fuel consumption at toll plazas, and supports sustainable infrastructure development. These features make it the ideal choice over competing solutions, offering a unique blend of technological advancement and societal benefit. Choosing this solution aligns with the PLUS company's goals of improving revenue, lowering operational costs, and delivering a superior user experience.

The proposed AI-powered Intelligent Automated Toll Collection System significantly aligns with Malaysia’s commitment to achieving Sustainable Development Goals (SDGs).

Firstly, SDG 9: Industry, Innovation, and Infrastructure emphasizes the development of sustainable and resilient infrastructure (UnitedNations, 2024). By adopting cutting-edge technologies like AI and IoT, this system revolutionizes Malaysia's toll infrastructure, positioning the country as a global leader in smart, innovative solutions. Its ability to leverage these technologies enhances Malaysia's global competitiveness and sets a benchmark for future intelligent public infrastructure.

Secondly, SDG 11: Sustainable Cities and Communities focuses on creating more livable, efficient urban environments (UnitedNations, 2024). The automated toll system contributes directly to this goal by

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reducing traffic congestion and enhancing transportation efficiency. It minimizes waiting times at toll plazas, leading to cleaner urban mobility and a reduced carbon footprint from idling vehicles, promoting a more sustainable urban lifestyle (Shuhua Xy && Chuanwang Sun && Nian Liu, 2024).

Thirdly, SDG 12: Responsible Consumption and Production highlights the importance of reducing resource use and waste (UnitedNations, 2024). The proposed system minimizes the need for extensive physical infrastructure and manual processes, reducing material consumption and operational waste. Real-time traffic monitoring and congestion management decrease vehicle emissions by preventing idling, ensuring the responsible use of resources and energy (Shuhua Xy && Chuanwang Sun && Nian Liu, 2024).

In conclusion, this system not only advances traffic management but also supports Malaysia's journey toward sustainable development. Its innovative application of AI and IoT fosters scalable, eco-friendly, and efficient solutions, significantly contributing to the global push for smarter infrastructure, sustainable cities, and responsible resource consumption.

MILESTONES

To ensure successful and timely completion of final prototype, Incremental Development Model has chosen to be adopted in for this project. This approach involves breaking down the system into multiple smaller standalone manageable modules, each developed and tested incrementally. The schedule is designed to prioritise core functionalities while allowing for continuous improvement and integration of additional features over time. (Aniket Nagargoje && Pavan K Kankar && Prashant Kumar Jain && Puneet Tandon, 2021)

Incremental Development Model has been chosen due to its modular and efficient approach. After dividing the project into smaller increments, focused attention can be given to each module, ensuring high quality and functionality. Next, it allows for faster deliverables by enabling early delivery of working modules to demonstrate progress and functionality to stakeholders. Additionally, it offers flexibility because there will be not having any disruption caused to the entire project even if there are new changes and latest feedbacks. Subsequently, it mitigates risks by allowing potential issues to be identified and addressed early through completion and testing of individual modules. Lastly, testing is conducted incrementally on smaller sections of the system. This contributes to faster and more thorough debugging.

The development model of the AI-Powered Intelligent Automated Toll Collection System will follow a structured approach to ensure timely and efficient completion. The key milestones are outlined below.

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Table 6.1: Milestones

Activities	**Expected Outcome**	**Completion Date**
Project Title Approval	Proposed project titles to get approval from the project supervisor.	25/11/2024
Project Planning	Identified problems, solutions, and target markets for the project. Defined main modules of the project based on the identified solutions Set up project timeline and milestones	13/12/2024
Chapter 1: Introduction	Determined objectives, backgrounds, and target audiences for the project.	16/02/2025
Chapter 2: Research Background	Conducted background research related to PLUS Malaysia.	07/03/2025
Chapter 3: Methodology and Requirements Analysis	Identified existing technologies and methodologies relevant.	23/03/2025
Chapter 4: System Design	Designed system architecture of the project in a flexible and modular manner to allow for easy adjustments and integration as the project proceeds.	06/04/2025
System Development	Developed the system using related programming languages, libraries, frameworks, and toolkits.	20/04/2025

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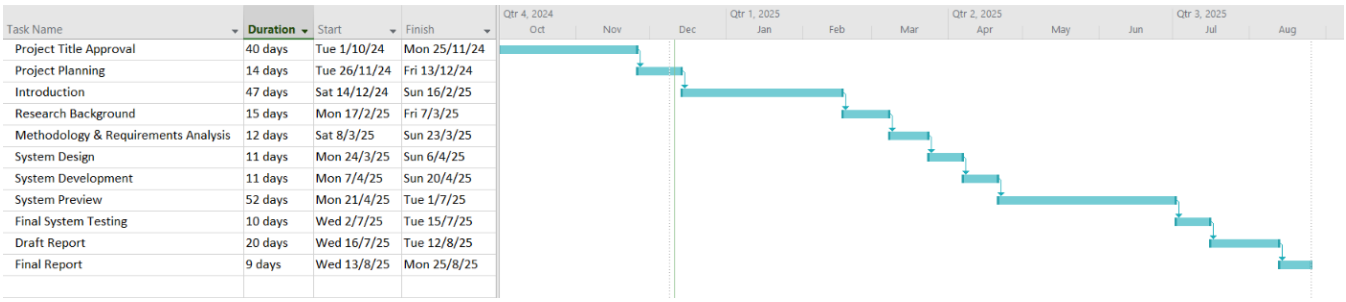
	Performed unit testing, integration testing, and system testing to identify and fix bugs inside the programs.	
System Preview	Demonstrated initial version of this project to supervisor and moderator.	01/07/2025
Final System Testing	Applied necessary testing on the fully integrated system, including performance testing, usability testing, and security testing to ensure that final system meets all requirements.	15/07/2025
Draft Report	Documented entire project process, including system design, development, testing, and user feedbacks.	12/08/2025
Final Report	Completed and organized all necessary components for submission.	25/08/2025

This schedule provides a well-organized roadmap, ensuring milestones are achieved progressively while maintaining focus on deadlines.



Task Name	Duration	Start	Finish
Project Title Approval	40 days	Tue 1/10/24	Mon 25/11/24
Project Planning	14 days	Tue 26/11/24	Fri 13/12/24
Introduction	47 days	Sat 14/12/24	Sun 16/2/25
Research Background	15 days	Mon 17/2/25	Fri 7/3/25
Methodology & Requirements Analysis	12 days	Sat 8/3/25	Sun 23/3/25
System Design	11 days	Mon 24/3/25	Sun 6/4/25
System Development	11 days	Mon 7/4/25	Sun 20/4/25
System Preview	52 days	Mon 21/4/25	Tue 1/7/25
Final System Testing	10 days	Wed 2/7/25	Tue 15/7/25
Draft Report	20 days	Wed 16/7/25	Tue 12/8/25
Final Report	9 days	Wed 13/8/25	Mon 25/8/25

Figure 6.1: Gantt Chart

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Figure 6.1 is the Gantt chart illustrating the milestones and timeline for completing the project. It provides a visual representation of the planned activities, their expected outcomes, and their respective deadlines, ensuring a clear overview of the project's progression.

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